



Problem solving with fractions and decimals

Task A: Solving problems with fractions

1) Arrange five different digits into these inequalities.

a) $\square > \frac{\square}{\square} > \frac{\square}{\square}$

b) $\frac{\square}{\square} > \frac{\square}{\square} > \square$

2) Arrange the digits 1 to 9 to make these inequalities true.

a) $\square > \frac{\square}{\square} > \frac{\square}{\square} > \frac{\square}{\square} > \frac{\square}{\square}$

b) $\frac{\square}{\square} > \frac{\square}{\square} > \frac{\square}{\square} > \frac{\square}{\square} > \square$

3) Izzy, Andeep, Lucas, Alex and Sam record their scores for a spelling competition.

Rank them from worst to best.

Izzy $\frac{11}{12}$

Andeep $\frac{4}{5}$

Lucas $\frac{5}{6}$

Alex $\frac{18}{20}$

Sam $\frac{26}{30}$



Task B: Solving problems with decimals

1) Here are the prices of some items in a cafe.



- a) Write them in order from cheapest to most expensive.
- b) You have a £5.00 gift voucher but you won't get any change.
You therefore want to spend as close to £5.00 as possible.
What can you buy?
- c) What is the largest number of items you can buy with £5.00 and what are they?

Task C: Solving problems with fractions and decimals

- 1) Work out the length of ribbon needed for cushions with these dimensions. (All dimensions are in metres)

Width	Length	Converted length / width	Calculation	Length needed
0.64	$\frac{9}{20}$			
1.38	$\frac{3}{5}$			
0.15	$\frac{3}{25}$			
$\frac{7}{20}$	1.07			
$\frac{7}{8}$	0.97			

- 2) A stopwatch measures time in minutes and seconds:

20:50

- a) Is this more or less than 20.5 minutes?
By how much?

20:30

- b) Is this more or less than 20.3 minutes?
By how much?

- c) Which time is closer to $20\frac{5}{10}$ minutes?

- d) Which time is closer to $20\frac{1}{3}$ minutes?

- e) Which do you think are further apart: 20.4 minutes and $20\frac{1}{4}$ minutes, or 20.5 minutes and $20\frac{1}{5}$ minutes.